

Eol Template Outline (+/- 1,500-2,500 words per Eol) – Due date CoB September 19

Please submit Eols to Masud Rana (MASUD.RANA@cgiar.org)

Section 0 – Title and contact Information

- Title of the proposed solution track: **Market led partnerships for resilient agri-food systems in Terai region of Nepal**
- Title of the AoW2 Flagship that the solution track aligns with: **Scaling seed + fertilizer soil fertility management systems for resilient, diverse and nutrition smart farming**
- Lead Scientist(s) / Contact Person(s) / Center(s): **Dyutiman Choudhary, CIMMYT, Kathmandu, Nepal**
- Emails of the Lead Scientist(s) / Contact Person(s) / Center(s): d.choudhary@cgiar.org
- Key CGIAR Center Partners (in the case of multi-Center Eols): **Santosh Nepal, IWMI, Nepal, s.nepal@cgiar.org**
- Technical link to other Science Programs – name the corresponding Science Programs by name: Digital Accelerator, Sustainable Farming, Policy Innovation

Section 1 – Innovation Package, Value Proposition & Alignment

Purpose: To establish what the innovation is, who the customers and end-users are, why it matters, and how it connects to S4I's Flagships and Grand Challenges.

- **The problem to be solved:** Describe the challenge you are seeking to overcome through the proposed Scaling Solution track. What country / countries will the Solution Track operate in? Be sure to clarify who you are seeking to benefit with this work, with attention paid to gender and social inclusion. (~150 words)

There is a wide gap between the actual and potential yields of staple crops at the farm level in Nepal. A large part of this gap is a result of unorganized and inefficient input markets (seed, fertilizer, irrigation). Domestic seed production is small and insufficient to meet these requirements. Timely access to quality seed with locally relevant genetic traits (e.g., high yield potential, resilience to biotic and abiotic stresses, nutritional quality) and irrigation technologies during different crop stages is a gap in Nepal. This situation coupled with poor availability and high prices of fertilizer pose binding constraints to improving agricultural productivity and rural livelihoods in Nepal. Additionally, input market actors face low capacity and poor market development constraints to scale their business to tap unmet demand. Sustainable seed and integrated soil fertility management systems reduce yield gaps, raise productivity, and improve rural livelihoods while making agricultural markets in Nepal more efficient, inclusive, and resilient.

- **Your scaling ambition:** Describe the ambition of your proposed Scaling Solution track. Which of the KPIs do you believe you will contribute most towards, and at what level of quantitative and qualitative impact? (~150 words)

This track aims to scale proven seed, soil fertility management and irrigation technologies in rice-maize based farming systems in Nepal's Eastern and Western Terai. CIMMYT and its partners

will scale such technologies by making “**agri-input markets work for small holder farmers**” that support farmers, market actors, cooperatives, and facilitators such as NGOs and local government.

- About 237,500 HHs with 1.2 M persons (40% are women and youth), will be reached to produce rice, maize, legumes, and vegetables, in about 257,730 ha across 11 districts of Koshi, Lumbini and Sudurpaschim provinces.
- The intervention will benefit 800,000 consumers with quality food and will add value to 10,000 jobs in the ag inputs sector.
- The interventions will result in enhanced irrigation facilities across 10,000 ha of maize-producing land in Lumbini province, significantly boosting crop diversification and productivity.
- **Innovation & Value Proposition** – Describe the innovation or bundle or package (maturity, novelty, suitability for scaling) and the benefits it delivers to institutional partners/customers and end-users (e.g., productivity, resilience, nutrition, risk reduction, institutional capacity). Can include linkages to other science programs that will collaborate and work on the innovation/bundle/package (~150 words)

CIMMYT and its partners are promoting a ‘**maize commercial model (MCM)**’ that introduces maize in spring fallows after rice harvest to diversify cropping systems with new profitable maize, legumes, rapeseed and tubers for meeting food and nutrition needs of farmers with efficient market linkages. This led to the increased demand of seeds with tolerance to biotic and abiotic stress, early maturity and high yield of different crops grown in the sequence. The seed varieties are bundled with best management practices such as the 4Rs (right source, right rate, right time, right place) of nutrient management, improved irrigation scheduling and incorporation of N fixing & in-situ green manuring crops (mungbean, cowpea, soyabean, Sesbania, Sunhemp etc.), biochar and nutrient enriched composts to maintain optimum soil pH and soil organic matter. These technology bundles coupled with supportive policies and markets have led to a 20-40% increase in yield of rice, legumes, vegetables and about 100% for maize, leading to improved food security, household incomes and input business growth.

- **Customers & Alignment** – Identify the main institutional customers (ministries, donor programs, agribusinesses, lenders) and end-users (farmers, SMEs, cooperatives). Show how the track aligns to an AoW2 Flagship and Grand Challenge. (~100 words)

The main institutional customers of the MCM based farming system diversification are government institutions (federal and provincial ministries of agriculture, livestock, land management and cooperatives, and irrigation), local governments, agro-dealers, micro small and medium enterprises (MSMEs), cooperatives involved in agri-input management, and NGOs. The action is deeply aligned with the priorities of the World Bank and Asian Development Bank in boosting climate resilience, food security and jobs. By scaling seed, soil fertility management and irrigation systems through local MSMEs and extension agents the track contributes to the grand challenge to deliver inputs and services at scale through inclusive market systems.

- **Evidence of Impact** – Provide citations and evidence of both effectiveness (impact) and scalability in practice (uptake beyond pilots, adoption drivers). (~150 words)

Farmers in Nepal’s Eastern and Western Terai have increasingly adopted hybrid maize on rice-fallow lands, with 70% using hybrid seeds and 87% practicing line planting for better efficiency.

The introduction of seed drills cut sowing costs from US \$53 to \$26 per hectare, and machine planting was used by 29% of farmers. Best management practices, such as targeted urea application and crop pooling, led to a 15% reduction in production costs and improved climate resilience. From 2021 to 2025, spring maize adoption in the western Terai soared by 700%, covering 2,000 hectares, while average crop yields (rice, maize, legumes, vegetables) rose 21–85% over 2017 figures. Annual maize value of cooperatives reached US \$3.3 million which is an 850% increase since 2021). Additionally, IWMI and DWRI developed irrigation monitoring systems (canal flow, soil moisture) for the Babai irrigation area covering 36,000ha, enabling data-driven water management and supporting crop growth at critical stages (Nepal et al. 2025).

[NSAF Closeout report 210325.pdf](#)

[Pandit et al 2025 4Rs principle maize.pdf](#)

[NUE and Yield Tomato Soil Systems.pdf](#)

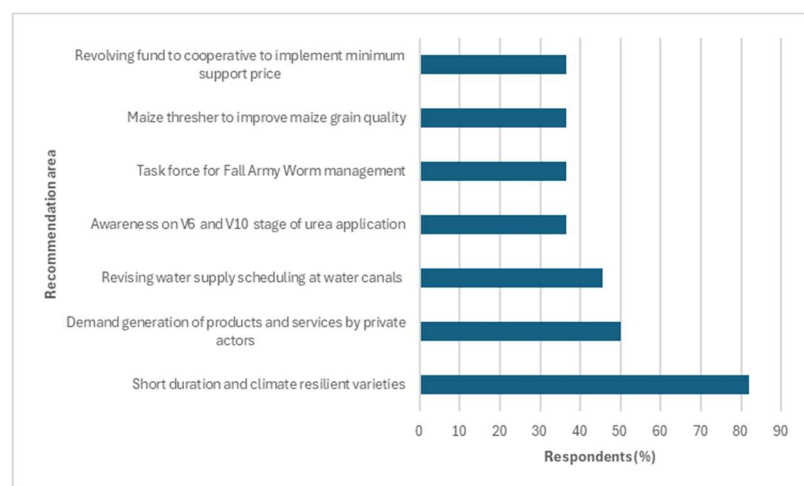
[Cereal Seed Value Chains in Nepal-2022.pdf](#)

<https://www.cimmyt.org/news/agro-fair-in-kailali-rejuvenates-farmers/>

Nepal, S.; K.C., J. K.; Khanal, A.; Neupane, N.; Bhatt, D. 2025. Basic instrumentation for informed decision-making in Nepal's large irrigation systems. Colombo, Sri Lanka: International Water Management Institute (IWMI). CGIAR Policy Innovations Program; CGIAR Scaling for Impact Program. 8p.

[Nepal pilots smart tools to transform its irrigation water management](#)

Fig 1 shows the rating provided to technology bundles during CIMMYT/IWMI Scaling Scan workshop for improving agri food system, with top priority for seed, irrigation and soil fertility management.



Source: Scaling scan workshop, May 2025, Bardiya Nepal

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

Purpose: To assess whether there is a credible pathway for the innovation to scale through public, private, or PPP delivery models — supported by sustainability plans, partner demand and commitments, financial leverage, and risk mitigation.

Scaling Pathway & Sustainability – Describe how the innovation bundle and/or package will scale (public, private, PPP). Explain how the solution will remain viable at scale (revenues, public financing, PPP, IFI/donor support), with milestones or cost-recovery targets. (~250 words)

The track will be implemented with a market-system approach with public-private-cooperatives partnerships that will scale seed and soil fertility pathways viz., i) delivery of quality seed (hybrids, OPVs, local landraces) and ii) delivery of soil fertility (extension, advisories) with soil amendments, seeds and irrigation inputs (equipment) to small holder farmers. Farmers will be mobilized through cooperatives for crop (rice, maize, legumes, vegetables, green manure) production and receive facilitation on cropping system optimization with climate resilient seeds, integrated soil fertility management (ISFM) practices and utilize digital seed system, soil maps and e-agrology. Input market actors such as seed dealers, farmer cooperatives, and facilitators such as local governments will be mobilized into multistakeholder platforms to assess demand and supply of inputs at the municipal level. These will create business cases for seeds, ISFM and irrigation products and services linking demand with supply. The market actors will develop input roadmaps and investment plans and will be trained to provide extension services, as part of their product sales and promotions, to farmers at various cropping stages. Local Governments will improve their agriculture strategies and provincial allocate budget to the tune of USD 0.5 million to support input provision for sustainable and resilient agriculture growth and market development. Solutions will sustain and remain viable as market actors and farmers will remain connected supported by technology, knowledge and policy support. Partnerships will be forged with output markets linked initiatives and improved production will provide surplus for household consumption and nutrition. Local Ag colleges will be mobilized for providing technical training to lead farmers, market and extension agents and to monitor the use and application of inputs and their impacts on improved crop production for feedback to local and provincial Governments. The system will also lead to a pool of trained human resources on input market development. The pathway will unlock new business opportunities with public private and donor funding of about USD 30 million.

Partnerships & Leverage – Identify internal CGIAR champions and external partners that have a stake in championing the innovation bundle or package, provide evidence of demand (surveys, requests for collaboration, alignment with policies or strategies) and commitment (MOUs, LOIs, budget allocations from partners, or other evidence of demand), and specify major non-CGIAR programs or investments this track can leverage (budgets, partners, policies, programs, etc). (~300 words)

The following partnerships and support will be leveraged:

- CIMMYT's network of partners – 15 seed companies, 33 seed cooperatives, 55 agrodealers & 50 local governments
- IWMI is working with the Department of Water Resources and Irrigation for effective water allocation for various crop in Babai irrigation system covering 36,000 ha and regularly monitors canal flow and soil moisture using sensors. The e-agrology system tested under the Digital Accelerator science program will be scaled for MEL and effective service delivery and tracking.

- The Department of Agriculture supported national soil mapping for the Digital Soil Map and testing services, and the Seed Quality Control Centre led digital seed information system will be utilized for information sharing and advisories.
- Partnerships will be developed with 11 Agriculture Knowledge Centers in 11 districts and 3 provinces (Lumbini, Sudurpaschim and Koshi) to harmonize provincial activity plans (e.g. block production program, output-based incentive, chief minister modernization program, marginalized community and youth focused programs) in line with market-based approach for seed and ISFM scaling
- Partnership with MoALD's rice mill model- This program is being implemented in 20 Terai districts (50 municipalities) and about 100 rice mills associated with 20,000 farmers are engaged in this program. There is potential for seed producers and input suppliers to sell 1,500 Mt of climate resilient, short duration and fine quality rice seed and apply ISFM technologies for rice production. After distributing seed, the millers will sign buyback agreements with farmers to ensure output market linkage.
- Partnership with MoALD seed and soil fertility programs through the Centre for Crop Development and Agrobiodiversity Conservation (CCDABC) that allocated USD 200,000 to 25 municipalities in FY 2025/26 for promotion and scaling climate resilient seed of multiple crops, fertilizer and irrigation innovations within and beyond MCM sites
- Strategic partnerships with district and province level Seed Entrepreneur Association and Fertilizer Association will be formed to ensure the timely planning, availability and distribution of quality seed and ISFM inputs.
- Partnership with the MoALD-JICA partnership “seed production and marketing” project being implemented in Eastern Terai districts (Jhapa, Morang and Sunsari) for promoting seed and ISFM technologies.
- Engagement with local Government Ag sector plans – local Government allocated USD 3.5 M in 2025 to support seed, irrigation and crop production in 50 municipalities in MCM areas.
- Agriculture and Forestry University in Koshi province and Far Western university in western Terai are actively engaged in mobilizing intern students for technology demonstration, awareness raising and implementing “learning and earning” programs which could be linked to this initiative
- **Risks & Bottlenecks** – Identify specific risks (financial, political, operational, social, or others) and bottlenecks (policy, institutions, finance, technology) in the specific context and place that could hinder scaling. Describe potential mitigation strategies to address these risks and bottlenecks. (*~250 words*)

The following risks are envisaged:

- Climate related – increasing heat stress and drought may impact crop production and scaling.
- Lack of access to quality inputs – access to high quality, stress tolerant and short duration seed varieties could be a challenge if supplies from India and China reduces. Additionally, availability of legumes seeds and cover crops can be short in supply if demand increases.
- Lack of fertilizers – is a chronic problem in Nepal and with increasing global prices and supply chain disruptions timely availability in the right quantity could be a challenge.
- Farmers lack of awareness – may delay the purchase of quality seeds, and application of ISFM practices for balanced soil fertility management.

- Government finances - increasing geopolitical tensions and inflationary pressures may reduce Government investment in agriculture development
- Political instability is a common feature in Nepal where Governments change frequently leading to policy instability and slowing down of Government programs.

Mitigation strategies

- Cropping calendars – develop cropping calendars to match sowing and harvesting dates with weather forecasts to avoid potential stressful periods.
- Implement seed road maps – to support seed sector stakeholders to plan in advance seed requirements especially for climate resilient varieties.
- Implement roadmap for balanced soil fertility management – to balance the use of organic resources, mixed cropping, green manuring with chemical inputs.
- Launch awareness campaigns through social media and Government channels for seed, ISFM and irrigation.
- Leverage investment from private sources for seed and ISFM value chain financing.
- Establish strong local stakeholder networks for PPP based interventions to overcome political instability.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

Purpose: To define what can realistically be achieved with S4I support in the next 2–3 years — including achievable reach, milestones, resources, activities, and safeguards for responsible scaling.

- **Readiness & Responsible Scaling** – Provide evidence of scaling readiness (IPSR or similar evidence). Describe actions for ensuring that scaling will be inclusive (extending to women, youth, marginalized groups) and environmentally sustainable. (~150 words)

CIMMYT and IWMI conducted a scaling assessment with stakeholders using the *scaling scan tool* (www.scalingscan.org) to assess the scalability of the maize commercial model (MCM) a production to market system innovation initiated by CIMMYT and its partners. During the session stakeholders prioritized potential socio-technical innovation bundles from maize and other crops in the cropping system that could best scale the production of these crops.

<https://www.cgiar.org/news-events/news/scaling-maize-production-in-nepal-the-mcm-initiative>

The scaling ambition is 'By 2030 CIMMYT, IWMI, government agencies and private sector will scale use of improved seeds, ISFM and irrigation technologies to 257,730 ha of Lumbini, Sudurpaschim and Koshi provinces. This scaling ambition will benefit 234,300 smallholder farmers households in the three provinces through scaling pathways designed around market system approach and public private partnerships. Social and environmental responsibility checks show that the scaling process is socially equitable where men, women and youth from various caste and ethnicities collaborate and environmentally sustainable with the promotion of climate resilient practices. The pathways include women, youth and indigenous people leading to their profitable integration in crop value chains.

Achievable target, Resources & Activities – Define what can realistically be achieved in 2–3 years starting 2026 forward with S4I support and current/planned partnerships. Outline the key resources (staff, institutional, financial, technical) and activities (e.g., training, policy engagement, distribution) required. (~250 words)

The following component wise activities will be implemented by the track in the next 3 years

Component 1: farmer and stakeholder mobilization,

Activities will include developing and strengthening multi stakeholder platforms for input market system coordination; matching demand and supply; developing input delivery and distribution strategies. Road maps for seed (OPV, hybrids, local land race production/import, distribution and delivery) and balanced soil fertility management technologies the 4Rs (right source, right rate, right time, right place) of nutrient management, improved irrigation scheduling and incorporation of N fixing & in-situ green manuring crops (mungbean, cowpea, soyabean, Sesbania, Sunhemp etc.), biochar and nutrient enriched composts, soil health cards derived from the DSM will be prioritized. Irrigation water supply and equipment for efficient use and delivery at key crop stages will be prioritized. Agroclimatic zone wise cropping calendars will be developed. Farmer mobilization, use of digital platforms for extension, training on best management practices will be carried out. The activity will reach 234,350 farming households (937,400 individuals), covering 257,730 ha in 50 municipalities of 3 provinces.

Component 2: Input market development and delivery

Activities will include seed and ISFM inputs product delivery to farmers; last mile distribution systems with PPP; local seed systems for landraces and improved varieties of soyabean, legumes, rapeseed by engaging local seed production institutions. Market actors and extension workers will be trained in input delivery and extension; seed and fertilizer associations will be supported with new research and business insights for investments. Market actors will be linked to farmers cooperatives and large Ag development programs for supplying quality inputs throughout the cropping season. Farmers output market linkages will be strengthened thus creating a self-sustaining business model for ag inputs and outputs. The activity will reach 200 market actors, provide quality inputs to 234,350 farming households (937,400 individuals) and provide 732,000 Mt of nutritious food to 1.5 million consumers.

Component 3: Knowledge assisted public private-cooperative partnerships

Activities will include developing joint ag input market development strategies; co-financing modalities with donor programs; market assessments and monitoring of activities. Trade fairs, investment forums, policy dialogues will be organized to boost the delivery of quality ag inputs to farmers.

The above activities will produce USD 441 M in sales; generate evidence for 2 policy readjustments; 2 seed and soil fertility management investment proposals and 3 market development reports.

Key resources

Staff – the intervention will work with professional staff with specialization in market development; soil fertility management; irrigation. Field staff will be hired for facilitation. CIMMYT and IWMI will collaborate in this intervention and work closely with the Sustainable Farming, Policy Innovation and Digital Accelerator science programs and explore linkages with the Breeding for Tomorrow science program.

Financial – this intervention will require a budget of USD 1.5 million over the period of 3 years. This will include salaries, sub- grants to partners, training, travel and workshops.

- **Timeline, Milestones & Monitoring** – Provide an estimate of annual milestones for 2026–2028 (users reached, partnerships formed, policies adopted). Explain how progress will be monitored, lessons captured, and adaptations made. (~200 words)

SN	Milestones	2026	2027	2028	Total
1	Seed companies engaged (No.)	7	10	15	15
2	Seed cooperatives engaged (No.)	22	33	33	33
3	Agrodealers engaged (No.)	44	55	55	55
4	Local nurseries engaged (No.)	33	44	44	44
5	Rice millers engaged (No.)	33	44	55	55
6	Municipalities engaged (No.)	50	50	50	50
7	Irrigation area expanded (ha)	2,000	5,000	3,000	10,000
8	Beneficiary HHs covered (No.)	65,600	85,700	86,200	237,500
9	Estimated seed distribution (t)	1,499	1,969	1,969	5,447
10	Value of seed transactions (US\$) M	3.4	4.5	4.5	10.8
11	Estimated area covered by improved seed, fertilizer and irrigation (ha)	64,900	84,700	84,700	234,300
12	Total estimated production of grain (t)	665,225	868,175	868,175	2,401,575
13	Total quantity for home consumption (t)	202,813	264,688	264,688	732,188
14	Total sales of produce in the market (t)	462,413	603,488	603,488	1,669,388
15	Values of sales of produce (USD) M	122	159	159	441
16	Average income from produce (US\$/HH)	1,860	1,860	1,860	5,580

17	Policy adjustments			2	
18	Investment proposals			2	
19	Market development reports			3	

We will use business performance monitoring of seed companies, agro dealers, service providers, random village surveys in combination with special studies that characterize the technology use and performance in farmers' fields. A framework for monitoring the MCM is being developed in 2025 which will be applied to determine input use. We will employ credible inference techniques to estimate both the uptake and impacts associated with different technologies.

Optional Annexes

1. Scaling Scan results

During the scaling scan exercise CIMMYT and IWMI solicited preferences for technology bundles from stakeholders. The innovations at various levels of the crop value chain that could be bundled were identified by the stakeholders and are shown in table 1. Table 2 shows the perception of use of technologies in their respective areas, where the application of innovations is found to be higher, indicating greater potential of scaling.

Table 1. List of socio-technical innovations identified by stakeholders

SN	Socio-technical Innovations	Input supply	Crop production	Harvesting	Aggregation and marketing
1	Joint action planning with stakeholders				
2	Agri-input fair at community level				
3	Selection of seeds of maize variety/hybrids based on cropping system requirement				
4	Demand aggregation and supply management by cooperative				
5	Preparation of biochar from weeds				
6	Demand generation of inputs by seed company, CHCs, agrodealer through field meetings and demonstrations				

7	Soil health cards to optimize the fertilizer types and amount				
8	Maize-legume intercropping				
9	Maize planting using precision maize planter				
10	Maize planting using drum seeder/ Jab planter				
11	Mobilizing digital champion to promote the use of digital advisories				
12	Digital crop insurance				
13	Formation and mobilization of task force for FAW management				
14	Mechanized Weeding by machine				
15	V6 and V10 stage of urea application				
16	Irrigation scheduling in negotiation with water user association (WUA)				
17	Time of harvesting using harvesting index				
18	Rouging of diseased and poor-quality plants from field				
19	Cob sorting before threshing				
20	Storage of matured plants with cobs at shade				
21	Buyback agreement process with millers				
22	Cooperatives' visit to feed mills				
23	Lending scheme from cooperatives				
24	Maize commercial network meeting for co-creation and fast-track communication				

25	Use of small electric thresher in community on rotation basis				
26	Multi-function thresher to improve grain quality				
27	Maize grain aggregation at cooperative for marketing				

Table 2. Stakeholders' perception of status of innovation use in the community (% respondents, n = 26)

SN	Innovations	Not aware	Testing	Little use (<10% area coverage)	Widely use (>10% area coverage)	Total use
1	V6 and V10 stage of urea application in maize	9.1	4.5	36.4	50.0	86.4
2	Digital champion to promote digital agro-advisory	4.5	9.1	54.5	31.8	86.4
3	Time of harvesting using harvesting index	4.5	9.1	54.5	31.8	86.4
4	Joint action planning with stakeholders		13.6	18.2	68.2	86.4
5	Demand generation strategy for CHCs, agrodealer by conducting training cum demonstration activities	4.5	13.6	36.4	45.5	81.8
6	Demand aggregation and input distribution by cooperatives		18.2	59.1	22.7	81.8
7	Removal of diseased cobs	9.1	9.1	31.8	50.0	81.8
8	Use of electric threshers at household/community level	9.1	9.1	40.9	40.9	81.8

9	Maize planting using precision maize planter	9.1	9.1	50.0	31.8	81.8
10	Agri-input fair at community level	4.5	18.2	36.4	40.9	77.3
11	Selection of seeds of maize variety/hybrids based on cropping system requirement	4.5	18.2	36.4	40.9	77.3
12	Rogueing of diseased plants from field	9.1	13.6	63.6	13.6	77.3
13	Mechanized weeding by machine	4.5	18.2	68.2	9.1	77.3
14	Formation and mobilization of task force for FAW management	13.6	13.6	50.0	22.7	72.7
15	Soil health card for soil health monitoring and optimize fertilizer	13.6	13.6	45.5	27.3	72.7
16	Cooperative level aggregation for marketing	9.1	18.2	59.1	13.6	72.7
17	Storage of matured plants with cobs at shade	22.7	13.6	40.9	22.7	63.6
18	Buyback agreement process with feed mills	4.5	31.8	45.5	18.2	63.6
19	Cooperatives visit to feed mills	27.3	9.1	45.5	18.2	63.6
20	Maize commercial network for Joint planning, sharing, monitoring policy lobbying	22.7	13.6	45.5	18.2	63.6
21	Irrigation scheduling in negotiation with water user association (WUA)	18.2	18.2	50.0	13.6	63.6
22	Special loan product by cooperatives for marginalized groups, youth	31.8	9.1	54.5	4.5	59.1
23	Digital crop insurance	13.6	31.8	45.5	9.1	54.5

24	Maize-legume intercropping	18.2	31.8	36.4	13.6	50.0
25	Use of multi-purpose thresher at commercial maize production	22.7	31.8	18.2	27.3	45.5
26	Biochar from weeds	18.2	50	27.3	4.5	31.8
27	Force fan for drying harvested cobs	36.4	36.4	22.7	4.5	27.3

2. IPSR Innovation Profile - Irrigation

An innovative irrigation instrumentation system is being piloted in the Babai scheme (36,000 ha) in western Nepal. Since 2024, IWMI and the Department of Water Resources and Irrigation have implemented this initiative, monitoring canal flow and soil moisture. An irrigation dashboard supports irrigation managers with data-driven insights for more effective water allocation and planning. The detailed IPSR is being planned in September/October 2025. Based on the IPSR readiness calculator, the scaling readiness is 6.

3. MELIA studies, impact assessments, reports, publications

- Khanal, N.P.; Choudhary D.; and Thapa, G. (2024). Enhancing food security in the era of rising fertilizer prices: evaluation of an intervention promoting mungbean adoption in Nepal. Asian Development Review 42 (1-29). <https://doi.org/10.1142/S0116110525500015>
- Ghimire, T.B.; Thapa, M.; Humagain, R.; Khanal, N.P; Choudhary, D.; Shrestha, H.K.; Gautam, S.; and KC, D. (2020). Status of National Seed Vision Implementation and issues. Kathmandu, CIMMYT International. <https://repository.cimmyt.org/server/api/core/bitstreams/729a05a0-04a3-43a8-a22f-2472f1308368/content>
- Thapa, G.; Choudhary, D.; Khanal, N.P.; and Gautam, S. (2023). Can traders' meetings boost seed sales of new rice varieties and promote a market driven seed system? Evidence from Nepal. Journal of Agribusiness in a Developing and Emerging Economies. DOI 10.1108/JADEE-09-2022-0193. <https://doi.org/10.1108/JADEE-09-2022-0193>
- Choudhary, D.; Banskota, K.; Khanal, N.P.; McDonald, A.J.; Krupnik, T.J.; and Erenstein, O. (2022). Rice subsector development and farmer efficiency in Nepal: Implications for further transformation and food security. Front. Sustain. Food Syst. doi.org/10.3389/fsufs.2021.740546

- Choudhary, D.; Banskota, K.; Gnawali, P.; and Khanal, N. P. (2019). Role of access to finance for small holder's seed business growth in Nepal: An Assessment. Development in Practice. <https://doi.org/10.1080/09614524.2020.1828828>
- Choudhary, D.; Khanal, N.P.; Gautam, S.; Shrestha, H.K.; KC, D.; and Beshir, A. (2020). Building a viable maize hybrid value chain in Nepal: Recent successes and the road ahead. Enterprise Development and Microfinance. <https://doi.org/10.59552/nppr.v3i1.59>
- Gautam, S.; Rahut, D.B.; Guzman, D.B.; Dongol, P.; KC, D.; Beshir, A.; and Choudhary, D. (2024). Does subsidizing seed help farmers? Nepal's rice seed subsidies. Dev Policy Rev. <https://doi.org/10.1111/dpr.12802>
- Gautam, S.; Dongol, P.; Khanal, N.P.; Beshir, A.; and Choudhary, D. (2023). COVID-19 disruption in agricultural input supply and impact on food security in Nepal. Trop. Agric. (Trinidad) Vol 100 No. 3 July 2023.
- Choudhary, D.; Khanal, N.P.; KC, D.; Gautam, S.; Shrestha, H.K.; and Beshir, A. (2018). Maize Market Development in Nepal: Challenges and Prospects. In: BM Prasanna, Aparna Das and Kelah K. Kaimenyi (editors). 2018. Book of Extended Food, Feed, Nutrition and Environmental Security. Ludhiana, India, October 8-10, 2018. CIMMYT, Mexico D.F.
- Gautam, S.; Joshi, N.K.; KC, D.; Choudhary, D.; Beshir, A.; Khanal, N.P.; and Shrestha, H.K. (2018). Technical Efficiency of Smallholder Maize Growers in Nepal and its Implications on Extension Services. In: BM Prasanna, Aparna Das and Kelah K. Kaimenyi (editors). 2018. Book of Extended Summaries, 13th Asian Maize Conference and Expert Consultation on Maize for Food, Feed, Nutrition and Environmental Security. Ludhiana, India, October 8-10, 2018. CIMMYT, Mexico D.F.
- Pandit, N.R.; Gaihre, Y.K.; Gautam, S.; Maharjan, S.; Vista, S.P.; and Choudhary, D. (2022). Enhanced-efficiency nitrogen fertilizer boosts cauliflower productivity and farmers' income: Multi-location and multi-year field trials across Nepal. Experimental Agriculture. [doi:10.1017/S0014479722000060](https://doi.org/10.1017/S0014479722000060)

4. Media reports and Blogs

- [Nepal uses digital soil map for the first time in SAARC](#)
- [धान, मकै र गहुँमा हाले मलको मात्रा तोकियो, कहाँ कति हाल्नुपर्छ? :: सुनिता सिटौला :: Setopati](#)
- [From seed to feed – CIMMYT](#)
- [Nepal Government endorses new site-specific fertilizer recommendations for rice – CIMMYT](#)
- [Balanced fertilizer application boosts smallholder incomes – CIMMYT](#)
- [Driving innovation through digital agriculture: Digital Ag Nepal, 2024 – CIMMYT](#)

5. Partner commitments (MOUs, LOIs, letters of support)

Below are the letters of support received from various Government agencies that support research and scaling.

1. [250107 Acknowledgement letter from DWRI.pdf](#)
2. [250326_letter_of_appreciation_from_PMAMP.pdf](#)
3. [DoA.pdf](#)
4. [MoALD.pdf](#)
5. [NARC.pdf](#)
6. [SQCC.pdf](#)

Annex 1: Resources & Tools to Support EOI Preparation

Teams preparing EOIs are encouraged to draw on the following core resources. These tools are directly aligned with the S4I operational criteria and will help ensure EOIs are evidence-based and robust:

Core Resources

1. Scaling Scan

- Practical tool to assess strengths and weaknesses across 10 “scaling ingredients.”
- Helps identify gaps in demand, finance, governance, partnerships, or capacities.
- Resource: [Scaling Scan](#)

2. Innovation Packages and Scaling Readiness (IPSR)

- CGIAR’s framework for assessing innovation maturity (readiness) and use, and for identifying bottlenecks and strategies for scaling.
- Can provide a Scaling Readiness Level (SRL) and bottleneck map useful for Criterion 3.
- Resource: [CGIAR IPSR Overview](#)

2-3. Business Model Canvas (BMC-lite)

- Adapted version of the BMC for development contexts.
- Useful for thinking through value proposition, partnerships, cost structures, and sustainability.
- Resource: [Business Model Canvas](#): Templates and examples available via AoW2 co-leads.

3-4. Responsible Scaling Guidance

- [GenderUp](#): A tool to integrate gender and inclusion into scaling design, helping identify trade-offs and opportunities for equitable, responsible scaling
- Helps applicants strengthen Criterion 3 (Responsible Scaling safeguards).

4-5. MELIA (Monitoring, Evaluation, Learning, Impact Assessment) Evidence

- [Past MELIA studies](#), adoption surveys, or bilateral project evaluations are valuable sources of impact evidence.
- Resource: Contact MELIA focal points within your Center for relevant reports and datasets.